Hyungjun Doh

Email: hdoh@purdue.edu **LinkedIn**: hyungjun-doh99 **Website**: danieldoh.github.io **Phone**: (765)-491-1304

EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Computer Engineering August 2018 – Present
GPA: 3.64 / 4.00

RESEARCH EXPERIENCE

Undergraduate Research Assistant January 2023 – Present
Advisor: Prof. Karthik Ramani Purdue University, West Lafayette
Convergence Design Lab

An Exploratory Study on Multi-modal Generative AI in AR Storytelling Lead

- Summarized a design space of multi-modal AR Storytelling and implemented a cognitive model for understanding the roles of authors and audiences in the storytelling process..
- Developed an experimental AR Storytelling platform with AI-generated multi-modal content, integrating with multiple state-of-the art generative AI models.
- Investigated the impact of AI-generated multi-modal content on the creation and perception of AR Storytelling through an exploratory study.

An HCI-Centric Survey and Taxonomy of Human-Generative-AI Interactions **[1]** Co-Author

- Reviewed 154 papers on Generative AI applications.
- Analyzed the recent advancements and research in the domain of GenAI-based systems.
- Contributed to synthesizing a taxonomy of human-GenAI interactions for future design in the field.

Visualizing Causality in Mixed Reality for Manual Task Learning: Co-Author
An Exploratory Study through Generative Artificial Intelligence **[2]**

- Implemented a MR interface to facilitate the learning of an assembly task by incorporating visual representations of causal relationships within the task.

CARING-AI: Towards Authoring Context-aware Augmented Reality Co-Author
INstruction through Generative Artificial Intelligence

- Implemented an AR interface for authoring AR instructions based on textual input describing tasks, avatars' trajectory, and directional vision.
- Contributed by developing various application scenarios for this system in diverse contexts.
- Developed a user interface for a user study and evaluated the system.

Summer Undergraduate Research Fellowship (SURF) May 2023 – August 2023
Advisor: Prof. Karthik Ramani Purdue University, West Lafayette
Engineering Undergraduate Research Office (EURO), Convergence Design Lab

- Implemented an animation generation pipeline with the integration of two diffusion models that leverage textual descriptions and pose skeletons to generate content for the storytelling.
- Presented a poster at the SURF Symposium, conveying the key insights and contributions.

Vertically Integrated Projects (VIP) August 2022 – May 2023
Advisor: Prof. Mohammad Reza Jahanshahi Purdue University, West Lafayette
VIP Smart Cities Research Team

- Implemented a semantic segmentation neural network to detect cracks and scratches.
- Generated 56 cracks and scratches datasets, each comprising 308 images, by utilizing the Houdini program.

TEACHING EXPERIENCE

Undergraduate Teaching Assistant August 2023 – Present
ECE 57000: Introduction of Artificial Intelligence

- Conducted weekly office hours to provide one-on-one assistance to students.
- Assisted in grading assignments and provided constructive feedback to students.

Workshop Teaching Assistant July 2023
Gifted Education Research and Resource Institute (GER2I)

- Managed and supervised high school students throughout the workshop
- Instructed students in the implementation of a controllable robot car using Arduino.

ENGINEERING PROJECTS

Smart Tripod August 2023 – Present
Facilitator, Senior Design Purdue University, West Lafayette

- Designed and implemented a Hand Gesture Recognition system on the Raspberry Pi, involved developing and fine-tuning machine learning models to recognize and interpret various hand gestures.
- Contributed to Human Object Detection system, communicating with Raspberry Pi regarding object location and orientation data.

Package Manager Application Programming Interface January 2023 – May 2023
Team Leader, ECE 46100 Purdue University, West Lafayette

- Developed a Package Manager API using TypeScript and Firebase.
- Implemented 17 features, totaling 2029 lines of code, to enhance functionality and user experience.
- Utilized Continuous Integration (CI) and Continuous Delivery (CD) pipelines for efficient development workflows.
- Designed the website, supporting API features, based on ADA compliance.
- Conducted security analysis to ensure API robustness and protect against vulnerabilities.

AWARDS

ECE Great Work Award Spring 2023
(Howard J. Heim Memorial Scholarship)
Summer Undergraduate Research Fellowship Summer 2023
Purdue Dean's List and Semester Honors Spring 2023
Purdue Dean's List and Semester Honors Fall 2022
Purdue Dean's List and Semester Honors Spring 2022
Purdue Dean's List and Semester Honors Fall 2021
Purdue Dean's List and Semester Honors Spring 2019

COMPETENCES

Programming Python, C, C++, C#, JavaScript, TypeScript
Platforms and Frameworks Pytorch, OpenCV, Unity, MATLAB Houdini, Firebase, Git, OpenGL
Languages English, Korean

LEADERSHIP

School Organization President May 2021 – May 2022
Korean-American Scientists and Engineers Association Young Generation

- Led a team of club officers and members, fostering a collaborative and motivated environment.
- Coordinated 5 academic events for STEM students by inviting professors and alumni.
- Received the "YG Events of the Year" award for organizing an event, inviting professors from various departments to share their experiences and foster relationships between professors and students.

Drill Instructor August 2019 – March 2021

Republic of Korea Army

- Led a team of 18 experienced drill instructors and trained approximately 2100+ army recruits.
- Organized all drill plans and ensured the safety protocols and guidelines.

PREPRINTS

[1] Shi, J.*, Jain, R.*, **Doh, H.**, Suzuki, R., & Ramani, K. (2023). An HCI-Centric Survey and Taxonomy of Human-Generative-AI Interactions. arXiv preprint arXiv:2310.07127.

[2] Jain, R.*, Shi, J.*, Benton, A., Rasheed, M., **Doh, H.**, Chidambaram, S., & Ramani, K. (2023). Visualizing Causality in Mixed Reality for Manual Task Learning: An Exploratory Study. arXiv preprint arXiv:2310.13167.

CITIZENSHIP

Republic of Korea